SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, UG Programmes CE,CSE,ECE,EEE,IT &ME are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204
Estd:1980

Regulation: R20	II / IV - B.Tech. I - Semester								
ELECTRICAL & ELECTRONICS ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 BS 2101	Numerical Methods & Vector Calculus	BS	3	3	0	0	30	70	100
B20 EC 2101	Electronic Devices and Circuits	PC	3	3	0	0	30	70	100
B20 EE 2101	Network Analysis	PC	3	3	0	0	30	70	100
B20 EE 2102	Electromagnetic Field Theory	PC	3	3	0	0	30	70	100
B20 EE 2103	Electrical Machines-I	PC	3	3	0	0	30	70	100
B20 EE 2104	Networks Laboratory	PC	1.5	0	0	3	15	35	50
B20 EE 2105	MATLAB Simulation Laboratory	PC	1.5	0	0	3	15	35	50
B20 EC 2105	Electronic Devices and Circuits Lab With Simulation	PC	1.5	0	0	3	15	35	50
B20 EE 2106	Solar Energy Systems Laboratory	SOC	2	0	0	4	--	50	50
B20 MC 2102	Professional Ethics and Human values	MC	0	2	0	0	--	--	--
		TOTAL	**21.5**	**17**	**0**	**13**	**195**	**505**	**700**

Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2101	**BS**	**3**	**--**	**--**	**3**	30	**70**	**3 Hrs.**

NUMERICAL METHODS & VECTOR CALCULUS

(Common to CE, CSE, EEE&IT)

Pre-requisites: Basic concepts of calculus.

Course Objectives: Students are expected to learn

1	Numerical methods to solve algebraic and transcendental equations and the concept of interpolation and its use for equally and unequally spaced data points,
2	Methods for numerical evaluation of integrals and for solving first order ODEs.
3	Concepts of double, triple integrals and its applications.
4	Concepts of Gradient, divergence, curl.
5	Vector integral theorems.

Course Out Comes: At the end of the course students will be able to

S. No	OUT COME	Knowledge Level
1	Determine a real root of an algebraic or transcendental equation. Fit an interpolation formula and perform interpolation for equally spaced and unequally spaced data.	K3
2	Evaluate numerically certain definite integrals. Solve a first order ordinary differential equation by Euler and RK methods.	K3
3	Evaluate double integrals and determine the areas.	K3
4	Evaluate triple integrals and determine the volumes.	K3
5	Find the gradient of a scalar function, divergence and curl of a vector function.	K3
6	Solve simple problems using vector integral theorems.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Solution of Algebraic and Transcendental Equations:** Introduction, Bisection method, Method of false position, Iteration method &Newton-Raphson method. **Interpolation:** Introduction, forward differences, backward differences, and Central differences. Differences of a polynomial, Newton's forward, and backward interpolation formulae. Interpolation with unequal intervals:Newton's divided difference and Lagrange interpolation formulae.
UNIT-II (10 Hrs)	**Numerical Integration and solution of Ordinary Differential equations:** Trapezoidal rule, Simpson's $1/3^{rd}$ rule. Solution of first order ordinary differential equations subjected to initial conditions by Taylor's method, Picard's method, Euler's method, Modified Euler's method and Fourth order Runge-Kutta method.
UNIT-III (12 Hrs)	**Multiple integrals:** Double and triple integrals, Change of order of integration. Change of variables, applications to finding Areas and Volumes.

UNIT-IV (10 Hrs)	**Vector differentiation:** Scalar and vector point functions, Vector Differentiation, Gradient, Directional derivative, Divergence, Curl, Scalar Potential.
UNIT-V (14Hrs)	**Vector Integration**: Line integral, Work done; Area, Surface and volume integrals, Vector integral theorems: Greens, Stokes, and Gauss Divergence theorems (without proof).

TEXTBOOKS:

1.	Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

REFERENCE BOOKS:

1.	Advanced Engineering Mathematics, by Erwin Kreyszig,Wiley
2.	Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
3.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
4.	Peter O' Neil, Advanced Engineering Mathematics, Cengage.
5.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2101	PC	3	--	--	3	30	70	3 Hrs

ELECTRONIC DEVICES AND CIRCUITS

(Common to ECE & EEE)

Course Objectives:

1.	To analyze the operating principles, characteristics and electrical parameters of Diode, BJT, JFET and MOSFET.
2.	To illustrate the concepts of biasing in BJT, JFET and MOSFET.
3.	To analyze amplifier circuits using equivalent circuits.
4.	To illustrate the application of diodes in rectifiers and regulated power supply.
5.	To analyze the frequency response of small signal amplifiers.

Course Outcomes: At the end of this course, the students will be able to

S.No	Outcome	KL
1.	Analyze the characteristics and operation of Diode, BJT.	K4
2.	Deduce the stability factors of different biasing circuits of BJT.	K3
3.	Analyze the characteristics and operation of JFET and MOSFET.	K4
4.	Design the small signal BJT single stage amplifiers.	K4

SYLLABUS

UNIT-I (12 Hrs)	**Semiconductor diode and its applications:** Potential variation in graded semiconductors, Mass action law, Open circuited PN junction, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, characteristics of Tunnel diode and Zener diode. Half wave, Full wave and Bridge Rectifiers with and without filters (inductor, capacitor, L-section and **π-**section).
UNIT-II (8 Hrs)	**Bipolar junction transistors:** Operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α,β,γ relation between α,β and γ early effect or base-width modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.
UNIT-III (8 Hrs)	**Transistor Biasing Circuits:** Need for biasing, The operating point, reasons for shifting of operating point, Bias stability, different types of biasing techniques, stabilization against variation in Ico , VBE, & β. Bias compensation, thermal runaway, thermal stability.

UNIT-IV (8 Hrs)	**Field Effect transistors:** JFET construction and its characteristics, pinch off voltage, FET as a VVR, FET small signal model, MOSFET and its characteristics.
UNIT-V (12 Hrs)	**Transistors at low and High frequencies:** Transistor hybrid model, H-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.

Text Books:	
1.	Integrated Electronics: Analog and Digital Circuits and Systems: Jacob Millman, C Halkias, Chetan D Parikh. McGraw – Hill.
2.	Electronic Devices and Circuits: N Salivahanan and Suresh Kumar, Third edition, TMH.
Reference Books:	
1.	Electronic Devices and Circuits Theory, Boylsted, 10th Edition, Pearson.
2.	Electronic Principles: Albert Paul Malvino, McGraw-Hill.
e-Resources:	
1.	https://nptel.ac.in/courses/117/103/117103063/
2.	https://www.iare.ac.in/sites/default/files/lecture_notes/IARE_ECE_EDC%20NOTES.pdf

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2101	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 hrs.**

NETWORK ANALYSIS

Course Objectives:

1.	Understand the concept types of sources and applications of network theorems for analysis of electrical networks.
2.	Understand the transient behavior of electrical networks with DC excitation and understand basics of magnetic circuit.
3.	Understand AC fundamentals and the behavior of RLC networks for sinusoidal excitations.
4.	Understand the concepts of balanced and unbalanced three-phase circuits and electrical resonance.
5.	Understand the concept of two-port network, network functions, poles and zeros.

Course Outcomes: At the end of this course, the students will be able to

S.No	Outcome	KL
1.	Understand electric circuit laws and **Apply** theorems to solve electrical networks.	K3
2.	**Analyze** the transient behavior of electrical networks using differential equation, and understand the concept of dot convention.	K4
3.	Understand AC fundamentals and **Solve** R, L, C network with sinusoidal excitation.	K3
4.	Understand the concept of electrical resonance and **also able to solve** three- phase circuits under balanced and unbalanced condition.	K3
5.	**Apply** two-port network analysis for devices like amplifiers, transmission lines and understand the concept of network functions, poles and zeros.	K3

SYLLABUS

UNIT-I (10 Hrs)	**DC Resistive Circuits:** Review topics - (Resistance parameter,Kirchhoff's laws,ohms law, Energy sources: Ideal, Non-ideal). **[No question should be set on Review topics]** Independent and dependent sources, Voltage and current division rule, Star-Delta transformation, source transformation, Mesh analysis and Nodal analysis, numerical problems. **Network Theorems:** Superposition, Thevenin's, Norton's, Max Power Transfer, Numerical problems.
UNIT-II (10 Hrs)	**DC Transient Analysis:** Inductance, Capacitance**,** Source free RL, RC and RLC Response, Evaluation of Initial and Final values, application of Unit-step Function to RL, RC and RLC Circuits. Concepts of Natural, Forced and Complete Response-using Differential equations. **Magnetic circuits**: concept of self and mutual inductance, Dot convention
UNIT-III (10 Hrs)	**Sinusoidal Steady State Analysis:** Definitions of terms associated with periodic functions: Time period, Angular velocity and frequency, RMS value, Average value, Phase angle, Phasor representation, Principal of Duality, numerical problems. Steady state analysis of R, L and C circuits. Power Factor – Real, Reactive, Apparent and Complex power, power triangle, Node and mesh analysis of AC networks, numerical problems.

UNIT-IV (10 Hrs)	**Resonance:** Series resonance and Parallel resonance, selectivity, bandwidth, quality factor – numerical problems. **Three phase circuits:** Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits.
UNIT-V (10 Hrs)	**Two port network parameters** – Z, Y, ABCD and hybrid parameters, relations between two port network parameters, series, parallel and cascade connection of two port networks, Numerical problems. **Network Functions**:Network Functions, Concept of Poles and Zeroes, Restriction of Poles and Zeroes for Driving point and transfer function.

Text Books:	
1.	Engineering Circuit Analysis, William Hayat Jr. and Jack E. Kemmerley, 8th Edition, McGraw Hill International Edition.
2.	Fundamentals of Electric circuits, K.Alexander and Matthew Sadiku, 6th Edition, McGraw Hill International Edition.
3.	Introduction to Modern Network Synthesis, Van Valkenburg; John Wiley

Reference Books:	
1.	Network Analysis and Synthesis, by Ravish R. Singh, McGraw Hill Education (1 July 2017).
2.	Schaum's Outlines Electric Circuits,Nahvi, Mahmood, Ph.D., Edminister, Joseph A,7th Edition, Mcgraw Hill.
3.	Network Analysis, 3rd Ed, A Sudhakar Shyammohan.SPalli Tata McGraw Hill Education Pvt Ltd.
4.	Circuit Theory Analysis and Synthesis.,7 th Edition 2014 by Abhijit Chakrabarthi,DhanpatRai&Co.

e-Resources:	
1.	https://nptel.ac.in/courses/117/106/117106108/
2.	https://nptel.ac.in/courses/108/105/108105159/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2102	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 hrs.**

ELECTROMAGNETIC FIELD THEORY

Course Objectives: Students will

1.	Study the basic laws and theorems that calculate electric field and forces.
2.	Study the concepts about the effects of electric fields on conductors, insulators and their boundary conditions.
3.	Acquire the knowledge of basic laws and theorems that calculate magnetic fields.
4.	Study the properties of magnetic materials, their boundary conditions, magnetic potential & energy.
5.	Understand Maxwell's equations and their application to Electromagnetic Wave Propagation

Course Outcomes: Students will be able to

S.No	Outcome	Knowledge Level
1.	**Compute** the electrostatic forces and electric field intensity for given charge configuration by applying Vector Calculus/Coulomb's / Gauss's law.	**K3**
2.	**Calculate** the Electric Potential, Energy and **discriminate** the conductors and dielectrics by its properties, Laplace's and Poisson's equations.	**K3**
3.	**Calculate** the Magnetic field intensity by **applying** Biot-Savart's/ Ampere's law	**K3**
4.	**Illustrate** the properties of magnetic materials and **find** the Magnetic potential and energy	**K4**
5.	**Derive** Maxwell's equations and apply them to **Analyze** the EM wave in different domains and compute average power density	**K3, K4**

SYLLABUS

UNIT-I (12 Hrs)	**Electrostatics** Rectangular, cylindrical and spherical coordinate systems, Coulomb's law and superposition principle, different types of charge configurations, electric flux, electric field intensity and electric flux density, electric field intensity and electric flux density due to different charge configurations, Gauss's law in integral form and point form in terms of D, applications of Gauss' law, Divergence theorem
UNIT-II (12 Hrs)	**Electric potential, Energy & Dielectrics** Electric potential, calculation of electric potential for given charge configuration, electrostatic energy, Electrostatic boundary conditions, basic properties of conductors in electrostatic fields, capacitance, Poisons and Laplace's equations, solutions of Laplace's equations for single variables, uniqueness theorem, electric dipoles, polarization of dielectrics.

UNIT-III (10 Hrs)	**Magneto statics** Biot-savart's law, determination of magnetic field intensity and magnetic flux density due to various steady current configurations, continuity equation, curl of H, Ampere's circuits law in integral and differential form, applications of Ampere's law, Stokes theorem.
UNIT-IV (10- Hrs)	**Magnetic Potential & Energy** Calculation of scalar and vector magnetic potential, magnetostatics boundary conditions. The magnetic dipole, magnetization, bound current, inductance and energy in magnetic fields, Lorentz force equation.
UNIT-V (10 Hrs)	**Time varying fields & Electromagnetic waves** Faraday's laws, Lenz's law, Maxwell's equations, modification of ampere's circuital law for time varying fields – displacement current and current density, the uniform plane wave, plane wave propagation, skin depth, the pointing vector, poynting theorem and power considerations.
Text Books:	
1.	Engineering electromagnetics by William H. Hayt, John A. Buck McGraw-Hill Publishing Co. 8th edition.
2.	Principles of Electromagnetics by Mathew N.O. Sadiku, Oxford; 4th edition.
Reference Books:	
1.	Introduction to electro dynamics by D.J. Griffiths, PHI Learning; 3rd Edition (2012).
e-Resources:	
1.	https://nptel.ac.in/courses/108/106/108106073/#

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2103	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 hrs.**

ELECTRICAL MACHINES-I

Course Objectives: Students will study

1.	Electro-mechanical energy conversion principle and Construction, Operation and different excitation methods of DC Machines.
2.	Characteristics of DC machines, armature reaction and commutation.
3.	Starting and Speed control methods of DC motors and testing of DC machines.
4.	Construction and operating principle of 1-Φ transformer, voltage regulation, phasor diagram, Equivalent circuit and tests.
5.	Construction and principle of Auto transformer & 3-Φ transformer and types, parallel operation and different types of connections of 3-Φ transformer.

Course Outcomes: Student will be able to

S.No	Outcome	KL
1.	**Apply** the concepts of electro mechanical energy conversion to understand the construction and working principle of DC machine.	K3
2.	**Analyze** the performance characteristics of DC Machines using EMF and torque equations and understand the concepts of armature reaction and commutation.	K4
3.	**Analyze** the performance of DC Machines using various tests and understand the starting and speed control methods	K4
4.	Understandthe construction and operating principle and **analyze** the performance of transformer using various tests and its characteristics using phasor diagram, equivalent circuit.	K4
5.	**Illustrate** the constructional details, connections, parallel operation and cooling methods of a 3-Φ transformer and understand the construction and operating principle of auto-transformer.	K3

SYLLABUS

UNIT-I (08 Hrs)	**Electromechanical energy conversion:** Basic principles of energy, force and torque in singly and multiply excited systems. Construction and working principle of DC machines and methods of excitation.
UNIT-II (10 Hrs)	**D.C. Machines** D.C generators-emf equation, armature reaction, commutation, compensating winding, characteristics of various types of generators, applications. D.C. motors- torque equation, D.C. shunt, series and compound motors– characteristics & applications
UNIT-III (12 Hrs)	**Starting, Speed control of DC Motors and Testing of DC Machines** Starting methods and speed control of D.C. motors. Testing of D.C machines - separation of losses, Direct and Regenerative methods to test D.C. machines. Swinburne's test, field's test.

UNIT-IV (12 Hrs)	**1-Φ Transformer:** Principle, construction and operation of 1-Φ transformers, phasor diagram, equivalent circuit, voltage regulation, losses and efficiency. Testing: open & short circuit tests, Sumpner's test.
UNIT-V (08 Hrs)	**Auto Transformer and 3-Φ transformer:** Autotransformers: Construction, principle, applications and comparison with two winding transformer. 3-Φ transformer: Construction, Types of connections and their comparative features, Parallel operation, Cooling methods.
Text Books:	
1	Kothari.D.P and Nagrath.I.J., "Electrical machines", McGraw Hill Education; 4 edition 2010).
2	Bimbhra.P.S, Electrical Machinery, Khanna Publishers, 2011.
Reference Books:	
1	Mg Say, Theory, "Performance & Design of A.C Machines", CBS publishers.
2	Irving L. Kosow, "Electrical Machines & Transformers", Prentice Hall; 2nd Revised edition 1990.
3	Hill Stephen, Chapman.j, "Electric Machinery Fundamentals", McGraw-Hill Higher Education, 4 edition (2004).
e-books:	
1	http://nptel.ac.in/courses/108105017/

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2104	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 hrs.**

NETWORKS LABORATORY

Course Objectives:

1.	To experimentally verify various theorems of circuit analysis.
2.	To learn parameters of two-port networks.
3.	To measure resonance phenomenon in electric Circuits.
4.	To measure parameters of iron cored inductor.
5.	To learn phasor concept of R-L-C Circuit

Course Outcomes:

S.No	Outcome	KL
1.	Apply the concept of theorems and analyze responses in D.C resistive network.	K3
2.	Justify resonance concept for series R-L-C circuit.	K5
3.	Determine parameters of two port networks.	K4 & K5
4.	Determine parameters of iron cored inductor.	K4
5.	Determine phasor values in series R-L-C circuit	K5

LIST OF EXPERIMETNS

1	Maximum Power Transfer Theorem.
2	Superposition Theorem.
3	Thevenin's Theorem.
4	Norton's Theorem
5	Reciprocity Theorem
6	Two Port Network Parameters
7	Series Resonance.
8	Ohm's Law and Resistance of filament Lamp
9	Parameters of choke-coil
10	R-L-C Phasor Diagram.
	Add-On Experiments
11.	Thevenin's Theorem using AC supply.
12.	Superposition Theorem using AC supply

Text Books:

1.	Circuit Theory: Analysis and Synthesis by A. Chakarbarati 2017 Reprint Edition
2.	Fundamentals of Electric Circuits 6th Edition by Charles K. Alexander and Matthew N. O. Sadiku
3.	Engineering Circuit Analysis 8th Edition William.H.Hayt

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2105	PC	--	--	3	1.5	15	35	3 hrs.

MATLAB SIMULATION LABORATORY

Course Objectives: Students will

S.NO	
1.	Gain a fair knowledge skill on the programming and simulation in MATLAB Software environment.
2.	Learn about the programming to solve linear and non-linear equations
3.	Learn to write programs to solve the electrical circuits
4.	Learn to simulate the electrical and electronics circuits for understanding steady state and transient behavior.
5.	Learn to simulate simple practical electrical systems.

Course Outcomes: Students will be able to

S.NO	Outcome	Knowledge Level
1.	**Apply** MATLAB Programming to solve simultaneous linear and nonlinear equation	K3
2.	**Apply** MATLAB Programming, simulation to solve differential equations and function minimization using optimization.	K3
3.	**Apply** MATLAB Programming to **Analyze**the series resonance and transient response of RLC network	K4
4.	**Apply** MATLAB simulation to obtainthe single-phase diode rectifier and DC Shunt motor characteristics.	K3
5.	**Analyze** the performance of Three phase circuits, Solar power system by MATLAB simulation.	K4

LIST OF PROGRAMS

	Introduction to MATLAB software
	PART-A: MATLAB PROGRAMMING
1	Basic Matrix Operations and Solution of simultaneous equations
2	Solution of simultaneous equations using Gauss Elimination method
3	Iterative solutions for non-linear equations
4	Solution of ordinary differential equation by Runge Kutta (RK) method
5	Function minimization for optimization.
6	Series RLC resonance
7	Transient response of a series RLC network
	PART-B: MATLAB SIMULINK
8	Transient responses of series RLC network
9	Steady state simulation of an electric circuit
10	Single phase diode bridge rectifier
11	Solution of non-linear differential equation
12	Three Phase Circuit with balanced and unbalanced load
13	Solar power system
14	DC shunt motor characteristics

Reference Books:

1.	Getting started with MATLAB by Rudra Pratap, Oxford University Press, 2005.
2	MATLAB and its Applications in Engineering by Rajkumar Bansal, Pearson Publishers, ISBN-10: 8131716813, 2009.
3.	Numerical Methods In Engineering With Matlab. JaanKiusalaas Cambridge university press, New York, 2005.
4.	Power System Analysis Haadi Saadat IInd edition, McGraw-Hill College1998

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2105	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs**

ELECTRONIC DEVICES AND CIRCUITS LAB WITH SIMULATION

(Common to ECE & EEE)

Course Objectives:

1.	To analyse the modelling, characteristics and electrical parameters of Diode, BJT, JFET and MOSFET.
2.	To illustrate the concepts of biasing in BJT, JFET and MOSFET.
3.	To illustrate the application of diode in rectifiers and regulated power supply.
4.	To analyze single stage amplifier circuits using equivalent circuits.

Course Outcomes: At the end of this course, the students will be able to

S.No	Outcome	KL
1.	Apply the concepts of different electronic devices to verify their characteristics and measure the important parameters.	**K3**
2.	Analyze the performance of rectifier circuits with and without filters.	**K4**
3.	Analyze the performance of BJT and FET amplifier circuits.	**K4**
4.	Simulation and Design of small electronic circuits using BJT and FET.	**K4**

HARDWARE

1	Study and Analyze V-I Characteristics of Semiconductor Diode (Ge & Si), LED and Zener Diode.
2	Determination of Ripple Factor and Regulation Characteristics of Half Wave and Full Wave Rectifier With and Without Filter.
3	Study and Analyze the Characteristics of BJT in CE Configuration and determination of h-parameters.
4	Study and Analyze the Characteristics of BJT in CB Configuration and determination of h-parameters.
5	Study and Analyze the JFET Characteristics.
6	Design of Biasing Circuits for BJT and FET.
7	Design of Common Emitter amplifier using BJT.
8	Frequency response of Common Source FET amplifier.

SOFTWARE

1.	Simulation of V-I Characteristics of Semiconductor Diode, LED and Zener Diode.
2.	Simulation of Regulation Characteristics of Zener Diode.
3.	Simulation of CC Amplifier.
4.	Simulation of JFET Characteristics.
5.	Simulation of BJT Characteristics in CE Configuration.
6.	Simulation of BJT Characteristics in CB Configuration.
7.	Simulation of Common Source JFET amplifier.
8.	Simulation of Tunnel Diode Characteristics.

Reference Books:	
1.	Lab manual
e-Resources:	
1.	http://vlabs.iitkgp.ac.in/be/#
2.	https://nptel.ac.in/courses/117/103/117103063/
3.	

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2106	**SOC**	**--**	**--**	**4**	**2**	**--**	**50**	**3 hrs.**

SOLAR ENERGY SYSTEMS LABORATORY

Course Objectives:

1.	To indulge skills to the students related to PV Panels erection, connections Solar energy extraction and Solar power Measurement.
2.	Familiarize equipment like cooker, street lights etc. working on Solar power
3.	Have an idea of roof top solar PV panels maintenance and testing.
4.	Students will learn to estimate and plan required number of PV panels for given domestic load.
5.	Students also acquire the knowledge on MPPT device and Net metering

Course Outcomes: Students are able to

S.No	Outcome	KL
1.	**Demonstrate** PV Panels erection, connections, Solar energy extraction and Solar power Measurement.	K4
2.	**Explain** the working principle of solar equipment like cooker, street lights etc.	K3
3.	**Installation of** roof top solar PV panels and understand their maintenance	K6
4.	**Plan** to estimate required number of PV panels for given domestic load	K6
5.	**Integrate** MPPT device and Net metering for the solar power	K6

LIST OF EXPERIMENTS

1.	Plot sun chart and locate the sun at your location for a given time of the day
2.	Connect solar panels in series & parallel and measure voltage and current.
3.	Measure intensity of solar radiation using Pyranometer and radiometers.
4.	Construct a street light connection using Solar PV panel (15W)
5.	Installation for OFF Grid mode of Solar panel connection.
6.	Installation of solar panels in a given space to extract maximum solar energy
7.	Installation for ON Grid mode of Solar panel connection.
8.	Verification of maximum power point tracking converter on solar system
9.	Roof top domestic solar power planning and estimation
10.	solar cooker dismantling and reassembling
11.	Maintenance and testing of solar panels & converter systems.
12.	Installation of Net-meter and Energy measurement

Reference Books:

1.	Solar energy utilization. G. D. Rai. 3rd Edition, Khanna Publishers, 1987.
2.	Solar Photovoltaics- Fundamentals, Technologies and applications, Chetan singh Solanki, 3rd edition, PHI learnigPvt.Ltd, 2015.
3.	Science and technology of photovoltaics, P.Jayarama Reddy, BS Pubmications, 2004.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2102	MC	2	--	--	--	--	--	--

PROFESSIONAL ETHICS AND HUMAN VALUES

Common to CE, EEE, ME, AIDS & CSBS

Course Objectives: On completing this course student will be able to

1	Create awareness on Engineering Ethics and Human Values.
2	Instill Moral and Social Values and Loyalty.
3	Appreciate the rights of others.

Course Outcomes: By the end of the course, the student should have the ability to:

S.No	Outcome	Knowledge Level
1	Identify and analyze an ethical issue in the subject matter under investigation or in a relevant field.	K3
2	Identify the multiple ethical interests at stake in a real-world situation or practice.	K2
3	Articulate what makes a particular course of action ethically defensible.	K2
4	Assess their own ethical values and the social context of problems.	K3
5	Identify ethical concerns in research and intellectual contexts, including academic integrity, use and citation of sources, the objective presentation of data, and the treatment of human subjects.	K2
6	Demonstrate knowledge of ethical values in non-classroom activities, such as service learning, internships, and field work.	K3
7	Integrate, synthesize, and apply knowledge of ethical dilemmas and resolutions in academic settings, including focused and interdisciplinary research.	K4

SYLLABUS

UNIT-I (8 Hrs)	**Human Values:** Morals, Values and Ethics-Integrity-Work Ethic-Service learning Civic Virtue Respect for others Living Peacefully Caring Sharing Honesty -Courage-Cooperation Commitment Empathy Self Confidence Character Spirituality..
UNIT-II (8 Hrs)	**Engineering Ethics:** Senses of 'Engineering Ethics-Variety of moral issued- Types of inquiry Moral dilemmas Moral autonomy- Kohlberg's theory- Gilligan's theory -Consensus and controversy Models of professional roles-Theories about right action-Self-interest -Customs and religion Uses of Ethical theories Valuing time Cooperation Commitment.,
UNIT-III (8 Hrs)	**Engineering as Social Experimentation:** Engineering As Social Experimentation- Framing the problem- Determining the facts codes of Ethics- Clarifying Concepts- Application issues Common Ground -General Principles- Utilitarian thinking respect for persons
UNIT-IV (8 Hrs)	**Engineers Responsibility for Safety and Risk:** Safety and risk Assessment of safety and risk. Risk benefit analysis and reducing risk- Safety and the Engineer-Designing for the safety- Intellectual Property rights(IPR).,

UNIT-V (8 Hrs)	**Global Issues:** Globalization- Cross-culture issues-Environmental Ethics- Computer Ethics Computers as the instrument of Unethical behavior Computers as the object of Unethical acts Autonomous Computers-Computer codes of Ethics- Weapons Development -Ethics and Research Analyzing Ethical Problems in research.
Text Books:	
1.	Engineering Ethics includes Human Values" by M.Govindarajan, S.Natarajan- and, V.S.Senthil Kumar-PHI Learning Pvt Ltd-2009.
2.	"Engineering Ethics" by Harris, Pritchard and Rabins, CENGAGE Learning, India Edition, 2009.
3.	"Ethics in Engineering" by Mike W. Martin and Roland Schinzinger - Tata McGraw-Hill-2003.
4.	"Professional Ethics and Morals" by Prof.A.R.Aryasri, DhanikotaSuyodhana-Maruthi Publications.
5.	"Professional Ethics and Human Values" by A.Alavudeen, R.Kalil Rahman and M.Jayakumaran-LaxmiPublications.
6.	"Professional Ethics and Human Values" by Prof.D.R.Kiran-
7.	"Indian Culture, Values and Professional Ethics" by PSR Murthy- BS Publication.
8.	Professional Ethics by R.Subramaniam - Oxford publications, New Delhi

Regulation: R20	II / IV - B.Tech. II - Semester								
ELECTRICAL & ELECTRONICS ENGINEERING									
SCHEME OF INSTRUCTION & EXAMINATION (With effect from 2020-21 admitted Batch onwards)									
Course Code	**Course Name**	**Category**	**Cr**	**L**	**T**	**P**	**Int. Marks**	**Ext. Marks**	**Total Marks**
B20 BS 2204	Complex Variables and Statistical Methods	BS	3	3	0	0	30	70	100
B20 EC 2201	Electronic Circuit Analysis and Design	PC	3	3	0	0	30	70	100
B20 EE 2201	Electrical Machines-II	PC	3	3	0	0	30	70	100
B20 EE 2202	Electrical Power Generation Transmission & Distribution	PC	3	3	0	0	30	70	100
B20 EE2203	Electrical Measurements & Instrumentation	PC	3	3	0	0	30	70	100
B20 EC 2206	Electronic Circuit Analysis And Design Lab With Simulation	PC	1.5	0	0	3	15	35	50
B20 EE 2204	Electrical Measurements & Instrumentation Laboratory	PC	1.5	0	0	3	15	35	50
B20 EE 2205	Electrical Machines-I Laboratory	PC	1.5	0	0	3	15	35	50
B20 EE 2206	Smart Systems Laboratory	SOC	2	0	0	4	--	50	50
B20 MC 2201	English Proficiency	MC	0	2	0	0	--	--	--
TOTAL			**21.5**	**17**	**0**	**13**	**195**	**505**	**700**

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B20BS2204	BS	3	--	--	3	30	70	3 Hrs.

COMPLEX VARIABLES AND STATISTICAL METHODS

COMMON TO CE & EEE

Pre-requisites: Basic concepts of Probability theory and Baye's Theorem

Course Objectives: Students are expected to

1	Learn the concept of Analytic function and its implications. Applications in Electrostaticsand fluid flow problems.
2	Learn the concepts in complex integration and evaluation of real definite integrals.
3	Formulate and solvelinear difference equations.
4	Learn important concepts of Z-transform and their use to solve linear difference equations.
5	Know the concepts of discrete and continuous random variables, learn a few importantdiscrete/ continuous probability distributions
6	Learn Concepts of Sampling theory and develop a framework for testing of hypothesis for getting inferences about Population Parameters.

Course Out Comes: At the end of the course students will be able to

S. No	OUTCOME	Knowledge Level
1	Comprehend the concept of Analytic function and apply in Electrostatics and Fluid dynamics	K3
2	Determine Laurent series of functions about isolated singularities, and determine residues. Use the residue theorem to evaluate certain real definite integrals.	K3
3	Formulate and solve linear difference equations.	K3
4	Use Z-transforms to solve linear difference equations with constant coefficients.	K3
5	Identify a random variable as discrete/continuous, find its expected value and also fit a probability distribution for a given frequency distribution.	K3
6	Decide the test applicable and apply it for giving inference about Population Parameter basedon sample statistic for some large samples and small samples.	K3

SYLLABUS

UNIT-I (12 Hrs.)	**Functions of a Complex Variable** Limit and continuity of a function of a complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, determine an analytic function based onthe knowledge of its real and imaginary parts, Milne-Thomson method, Applications ofanalytic function to flow problems, and in Electrostatics. Conformal mapping: thetransformations defined by $w = z+c$, $w = cz$, $w = 1/z$. The Bilinear transformation.
UNIT-II (10 Hrs.)	**Complex Integration:** Line integral, Cauchy's integral theorem, Cauchy's integral formula. Expansion of a function in a Taylor series, McLaren series and Laurent series. Types of singularities, Residues, Cauchy's residue theorem. Evaluation of real definite integrals -integration around unit circle (Theorems without proofs).

UNIT-III (14 Hrs)	**Difference equations and Z-transforms:** Formation of a difference equation, Rules for finding complimentary functionand particular integral for linear difference equations. Definition of Z- transform, some standard Z- transforms, properties, Shifting U_n to the right and to the left, transform of a function multiplied by n, initial value theorem and final value theorem(without proof), evaluation of inverse Z- transforms, convolution theorem (without proof), solution of linear difference equations by the use of Z-transforms.
UNIT-IV (10 Hrs.)	**Probability Distributions:** A brief review of random variables, Binomial, Poisson and Normal distributions, definitions of pmf/ pdf, notation, mean, variance, moment generating function. Fitting of Binomial or Poisson distributions for a given frequency distribution.
UNIT-V (12 Hrs.)	**Sampling theory and Testing of Hypothesis:** **Sampling theory:** Introduction, population and samples, level of significance, procedure testingof hypothesis. **Large samples:** Testing of hypothesis for single proportion and two proportions. **Small samples:** Degrees of freedom, Students' t- distribution, t-test for single mean, two means; Chi- squared distribution, test for goodness of a fit.
TEXTBOOKS:	
1.	Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43^{rd} Edition, Khanna Publishers.
2.	Probability and statistics for Engineers, Miller and Freund, 7^{th} edition, Pearson 2008.
REFERENCE BOOKS:	
1.	Fundamentals of Mathematical Statistics by S.C.Gupta and V.K.Kapoor, Sultan Chand & Sons Publishers
2.	Probability and statistics for Engineers and Scientists by Ronald E. Walpole, Sharon L. Myers and Keying Ye, 8^{th} edition, Pearson Education, 2007.
3.	Advanced Engineering Mathematics, by Erwin Kreyszig,Wiley
4.	Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5.	A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
6.	Advanced Engineering Mathematics, by H.K.Dass, S.ChandCompany.
7.	Higher Engineering Mathematics, by Dr. M.K.Venkatraman, the National Publishing Company

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2201	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ELECTRONIC CIRCUIT ANALYSIS AND DESIGN

(Common to ECE & EEE)

Course Objectives:

1.	Apply the concept of multistage amplifiers and analyze them.
2.	Illustrate the analysis and design of feedback amplifiers, power amplifiers and tunedamplifiers.
3.	Illustrate the principle of oscillator and analyze different types of sinusoidal oscillators
4.	Illustrate the concept and analyze applications of op-amp.

Course Outcomes: At the end of this course, the students will be able to

S.No	Outcome	KL
CO1	Outline the concepts of multistage amplifiers, feedback amplifiers, power amplifiers, tuned amplifiers, operational amplifiers and oscillators.	**K2**
CO2	Apply the concepts in the realization of practical circuits.	K3
CO3	Analyze and design practical electronic circuits using amplifiers, oscillators and operational amplifiers.	K4

SYLLABUS

UNIT-I (10 Hrs)	**Multistage Amplifiers** Transistor at high frequencies, CE short circuit current gain and concept of GainBandwidth product. BJT and FET RC coupled amplifiers at low and high frequencies.Frequency response and calculation of Bandwidth of Multistage Amplifiers.
UNIT-II (10 Hrs)	**Feedback Amplifiers** Concept of Feedback Amplifiers - Effect of Negative Feedback on the amplifierCharacteristics. Four feedback topologies, Method of analysis of Voltage Series, CurrentSeries, Voltage Shunt and Current Shunt feedback Amplifiers.
UNIT-III (8 Hrs)	**Sinusoidal Oscillators** Condition for oscillations and types of Oscillators – RC Oscillators: RC Phase Shift andWien bridge Oscillators. LC Oscillators: Hartley, Colpitts, Clapp, Tuned Collector andCrystal Oscillators.
UNIT-IV (10 Hrs)	**Power Amplifiers:** Classification of Power Amplifiers. Series fed, Transformer coupled class-A and class-BPower amplifiers. Push Pull Class-A, Class-B and Class-AB Power Amplifiers.Cross-over Distortion in Pure Class-B Power Amplifier and Class-AB Power Amplifier-Trickle Bias, Derating Factor and Heat Sinks – Complementary Push Pull Amplifier. **Tuned Voltage Amplifiers:** Analysis of Single tuned, Double tuned and Stagger Tuned Amplifiers with gain and Bandwidth Calculations.
UNIT-V (10 Hrs)	**Operational Amplifiers:** Concept of Differential Amplifier. Differential Amplifier supplied with a constant current source. Calculation of common mode rejection ratio. Block diagram and Ideal characteristics of an Op-Amp. Applications of Op-Amp: Inverting and Non-Inverting amplifiers, Integrator,

	Differentiator, Summing, Subtracting and Logarithmic Amplifiers.Definition and Measurement of OP-Amp Parameters.
Text Books:	
1.	Integrated electronics-Jacob Millman and Christos C. Halkias
2.	Electronic Devices and Circuits – Robert L Boylstead& Lewis Nashalsky, 2009, 10th edition, Pearson publications.
3.	Op-amps and Linear Integrated Circuits – Gayakwad
Reference Books:	
1.	Electronic Devices and Circuits by Salivahanan. Tata McGraw-HillPublication.
2.	Electronic Devices and Circuits by Sanjeev Gupta, 3rd Edition, Dhanpat RaiPublications
e-Resources:	
1.	https://www.electronics-tutorials.ws/oscillator/oscillators.html
2.	https://www.electronics-tutorials.ws/systems/negative-feedback.html
3.	https://www.uotechnology.edu.iq/dep-eee/lectures/3rd/Electronic/Analog%2520electronic/part2.pdf
4.	https://www.electronics-tutorials.ws/opamp/opamp_1.html

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2201	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

ELECTRICAL MACHINES-II

Course Objectives: Students will learn

1.	The concept of synchronous generator, EMF equation and air-gap MMF distribution.
2.	Construction of synchronous machine, voltage regulation, power angle characteristics of salient and non-salient pole synchronous generator.
3.	The operation and staring methods of synchronous motor, Synchronization ti infinite bus, operating characteristics, V-curves
4.	construction of three phase induction motor, speed-torque characteristics, circle diagram
5.	Starting and speed control of 3-φ induction motor, double revolving field theory, equivalent circuit, starting method of 1-φ induction motors.

Course Outcomes: Students will be able to

S.No	Outcome	Knowledge Level
1.	Illustrate the constructional features of AC rotating machines, MMF distribution in air-gap, Rotating Magnetic field and calculate generated voltage.	K3,K4
2.	Compute voltage regulation and analyze the power angle characteristics of salient and Non salient pole synchronous generator.	K3, K4
3.	Explain Parallel operation of synchronous generator and Analyze characteristics of salient pole synchronous motor and explain the starting methods of synchronous motor.	K3, K4
4.	Illustrate the operation of three phase induction motor and apply the slip-torque equations in Analyzing the performance of 3Φ Induction Motor.	K3, K4
5.	Demonstrate the starting and speed control of 3-Φ induction motor, operation of single-phase induction motor and its starting methods.	K3,K4

SYLLABUS

UNIT-I (08 Hrs)	**BASIC CONCEPTS IN AC ROTATING MACHINES:** Constructional features- cylindrical rotor, Salient pole rotor, single turn coil - active portion and overhang; full-pitch coils, concentrated winding, generated EMF, distributed winding, Harmonic content in distributed winding, Short pitched coils (Simple Problems), Air-gap MMF of Distributed AC windings-MMF space wave of single coil, MMF space wave of one phase of distributed winding, Production of Rotating Magnetic Field (only concept).

UNIT-II (12 Hrs)	**SYNCHRONOUS GENERATORS:** Circuit model, armature reaction, Determination of synchronous Reactance-OCC and SCC, voltage regulation, voltage regulation by EMF, MMF and ZPF methods, Salient pole Machine -two reaction theory, analysis of phasor diagram, power angle characteristics, determination of x_d and x_q, synchronizing power coefficient, synchronizing power(Simple Problems).
UNIT-III (08 Hrs)	**PARALLEL OPERATION OF ALTERNATORSand SYNCHRONOUS MOTORS:** Synchronizing to infinite bus, Parallel operation of Alternators (Only concepts).Operating principle, starting methods of synchronous motors, salient pole synchronous motor-Analysis of phasor diagram (simple problems),power flow equations, Operating characteristics of synchronous motor-variable load and constant excitation, V-curves, variable excitation and constant load.
UNIT-IV (12 Hrs)	**INDUCTION MACHINES** Construction, Types (squirrel cage and slip-ring), Equivalent circuit- Starting and Maximum Torque, Torque Slip Characteristics, Effect of parameter variation on torque speed characteristics (variation of rotor resistance), No-load and Blocked rotor test, Losses and Efficiency, Circle Diagram.
UNIT-V (10 Hrs)	**STARTING AND SPEED CONTROL OF 3-Φ INDUCTION MOTOR AND1-Φ INDUCTION MOTORS** Methods of starting, braking and speed control for 3-Φ induction motors (simple problems). Self-excited Induction Generator. 1-φ induction motors-constructional features, double revolving field theory, equivalent circuit, determination of parameters (only concept), Split phase starting methods & applications.
Text Books:	
1.	Nagrath & Kothari, “Electric Machines” TMH 5th edition
2.	PS Bhimbra, “Electrical Machinery”, Khanna Publishers. 7th edition.
Reference Books:	
1.	Fitzgerald & Kingsley, “Electric Machinery” McGraw Hill, 6th edition

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2202	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 hrs.**

ELECTRICAL POWER GENERATION, TRANSMISSION & DISTRIBUTION
(For EEE)

Course Objectives: Students will

1.	Understand the general arrangement, principles and components & their functions present in thermal, hydro, nuclear and gas power plants.
2.	Understand the load curves and different types of tariffs.
3.	Know the performance analysis of transmission lines
4.	Give emphasis on mechanical design of transmission line conductors and insulators.
5.	Study the different types of distribution systems

Course Outcomes: Student will be able to

S.No	Outcome	Knowledge Level
1.	**Describe** the power generation from different energy sources, tariffs and Economic aspects.	K3
2	**Apply** Kelvin's law and **analyze** different type's transmission and distribution networks.	K3, K4
3.	**Calculate** Inductance & Capacitance of transmission lines	K3
4.	**Determine** the performance of short, medium and long transmission lines.	K3
5.	**Explain** the mechanical and electrical design aspects of transmission system	K3

SYLLABUS

UNIT-I (9–Hrs)	**ELECTRIC POWER GENERATION & ECONOMICCONSIDERATIONS**: Layout of thermal, hydro, nuclear and gas power plants, brief description of various parts of different power plants. Load curves and associated definitions, load duration curves, different types of tariffs and examples.
UNIT-II (8–Hrs)	**POWER SUPPLY SYSTEMS& DISTRIBUTION SYSTEMS**: Transmission and distribution systems- D.C 2-wire and 3- wire systems, A.C single phase, three phase and 4-wire systems, comparison of copper efficiency. Primary and secondary distribution systems, concentrated & uniformly distributed loads on distributors fed at both ends, ring distributor, voltage drop and power loss calculation, Kelvin's law.
UNIT-III (9–Hrs)	**INDUCTANCE & CAPACITANCE CALCULATIONS**: Types of conductors, line parameters, calculation of inductance and capacitance of single and double circuit transmission lines, three phase lines with bundle conductors. Skin effect and proximity effect.
UNIT-IV (8- Hrs)	**PERFORMANCE OF TRANSMISSION LINES**: Generalized network constants and equivalent circuits of short, medium, long transmission line. Line performance: regulation and efficiency, Ferranti effect.
UNIT-V (8–Hrs)	**OVERHEAD LINE INSULATORS**: Types of insulators, potential distribution over a string of suspended insulators, methods of equalizing potential. Corona: phenomenon

	of corona, corona loss, concept of radio interference. **MECHANICAL DESIGN OF TRANSMISSION LINES**: Different types of tower, sag –tension calculations, sag template, string charts.

Text Books:	
1.	Wadhwa,C.L., " Electric Power Systems", New Age International Private Limited; Sixth edition (2010).
2.	Power System Analysis and Design by Dr. B.R Gupta S Chand &Company; 2005.
3.	Nagarath,I.J, and Kothari, D.P., "Power System Engineering", McGraw Hill Education; 2 edition (2007).
4.	"A Course in PowerSytems" by J.B Gupta, S.K. Kataria& Sons; 2013 edition.
Reference Books:	
1.	Burke James,J., "Power Distribution Engineering; Fundamentals and Applications " Marcel Dekker 1996.
2.	Grainger john, J. and Stevenson,Jr. W.D., "Power System Analysis", McGraw Hill,1994.
3.	Harder Edwin,I., "Fundamentals of Energy Production",JohnWiely and Sons,1982.
4.	Deshpande, M.V., "Elements of Electric Power Station Design",A.H Wheeler andCo.Allahabad,1979.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2203	PC	3	--	--	3	30	70	3 hrs.

ELECTRICAL MEASUREMENTS AND INSTRUMENTATION
(For EEE)

Course Objectives:

1.	To impart the knowledge of working of various meters and instruments used for the measurement of various electrical quantities.
2.	To select and determine the usage of the measuring instrument for various applications and projects.
3.	To select appropriate AC and DC Bridges for measuring resistance, inductance, and capacitance.
4.	To impart the knowledge of working of various instruments used for the measurement of non-electrical quantities.
5.	To impart the knowledge regarding operation of ADC, DAC and CRO.

Course Outcomes: Students able to

S.No	Outcome	Knowledge Level
1.	**Examine** the operation of moving coil, moving iron and dynamometer type instruments for measuring electrical quantities such as voltage, current, power, energy and power factor.	**K3, K4**
2.	**Analyze** the compensation techniques for accurate measurement and extension of range of measurement.	**K3, K4**
3.	**Analyze** the usage of different bridges for the measurement of Resistance, Capacitance, Inductance and Frequency.	**K3, K4**
4.	**Examine** the operation of different transducers for measuring non-electrical quantities such as displacement, pressure and temperature.	**K3, K4**
5.	**Interpret** the usage of CRO, ADC and DAC.	**K3**

SYLLABUS

UNIT-I (10 Hrs)	**Measuring Instruments** Classification, Error Analysis, Deflecting, Controlling and Damping torques, Moving Coil, Moving iron type instruments, Expression for the torque, Errors, and compensations. Extension of range using shunts and multipliers.
UNIT-II (10 Hrs)	**Measurement of Power and Energy** Dynamometer type wattmeter - Torque expression, Errors, Extension the range of Wattmeter; Energy meter- Torque expression, Errors. Power factor meter.
UNIT-III (12 Hrs)	**AC & DC Bridges** Methods of Measuring Low, Medium, and High Resistance, Kelvin's Double Bridge, Wheat Stone's Bridge, Loss of Charge Method, Methods of Measuring Inductance, Capacitance, and Frequency-Maxwell's Bridges, Anderson's Bridge, Hay's Bridge, Owen's Bridge, Schering Bridge, Wein's Bridge.

UNIT-IV (10 Hrs)	**Measurement of Non-Electrical Quantities** Transducers – Classification, Characteristics of Transducers. Measurement of Pressure - Strain Gauge, Piezo-Electric Transducers; Measurement of Displacement -Potentiometer, LVDT; Measurement of Temperature -Thermocouple.
UNIT-V (8 Hrs)	**ADC, DAC and CRO** ADC- Flash Type, Dual Slope Type, Counter Type, Successive Approximation Type. DAC- R2R ladder, Weighted Resistor Type.CRO- Calibration, Measurement of frequency and Voltage, Lissajous Patterns.

Text Books:

1.	Sawhney, .A.K., "A Course in Electrical & Electronic Measurement and Instrumentation," Dhanpat Rai & Company Private Limited, New Delhi, 18th Edition, 2007.
2.	Golding, E.W., and Widdis, F.C., "Electrical Measurements and Measuring Instruments," A H Wheeler & Company, Calcutta, 5th edition, 2003.

Reference Books:

1.	Doeblin, E.O., "Measurement Systems: Application And Design," 5th Edition, Tata Mc-Graw Hill Publishing Company Limited, New Delhi, 2004.
2.	Rangan, C.S., Sharma, G.R., Mani, V.S., "Instrumentation Devices and Systems," Tata McGraw-Hill Publishing Company, New Delhi, 2nd Edition, 2002.

e-Resources:

1.	https://swayam.gov.in/nd1_noc19_ee44/
2.	https://youtube.com/playlist?list=PLIzyJ2-aS_hu_0glcqNKXXujEXlw8fZ2W

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EC2206	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

ELECTRONIC CIRCUIT ANALYSIS AND DESIGN LAB WITH SIMULATION

(Common to ECE & EEE)

Course Objectives:

1.	To prepare students to perform the analysis of any Analog electronics circuit.
2.	To empower students to understand the design and working of BJT / FET amplifiers, oscillators and Operational Amplifiers.
3.	To evaluate the use of computer based analysis tools to review performance of semiconductor device circuits.
4.	Model the electronic circuits using tools such as PSPICE and Multisim.

Course Outcomes: On completion of this laboratory course, the students will be able to

S.NO	Outcome	KL
1.	Apply the concepts of amplifier analysis to verify their characteristics and measure the important parameters.	K3
2.	Analyze the performance of power amplifiers.	K4
3.	Analyze the frequency response and characteristics of operational amplifiers.	K4
4.	Simulation and Design of different amplifiers and oscillator circuits.	K4

LIST OF EXPERIMENTS

HARDWARE

1	Frequency Response of RC Coupled Amplifier.
2	Frequency Response of Negative Feedback Amplifier
3	Colpitts Oscillator.
4	RC Phase Shift Oscillator.
5	Wein Bridge Oscillator.
6	Hartley Oscillator.
7	Basic Applications of Operational Amplifier.
8	Tuned Voltage Amplifier.

SOFTWARE

1	Frequency Response of RC Coupled Amplifier.
2	Frequency Response of Negative Feedback Amplifier
3	Colpitts Oscillator.
4	RC Phase Shift Oscillator.
5	Wein Bridge Oscillator.
6	Hartley Oscillator.
7	Basic Applications of Operational Amplifier.
8	Tuned Voltage Amplifier.

Reference Books:

1. Lab manual.

e-Resources:

1. https://www.orcad.com/resources/orcad-tutorials

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2204	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

ELECTRICAL MEASUREMENTS AND INSTRUMENTATION LABORATORY

(For EEE)

Course Objectives:

1.	To identify the experimental setup for measuring resistance, inductance, capacitance, power, energy & dielectric strength.
2.	To demonstrate the quality of inductance & capacitance using AC bridges.
3.	To identify the efficient measurement and calibration of wattmeter.
4.	To understand the operating principle of transducers while measuring displacement & force.
5.	To identify the efficient measurement and calibration of energy meter.

Course Outcomes:

1.	**Determine** the quality factor of inductance using Anderson and Maxwell Bridge	**K4**
2.	**Examine** the necessity of calibration for wattmeter, energy meter, LVDT, Thermocouple and Strain Gauge.	**K4**
3.	**Analyze** the experimental setup for measuring resistance, capacitance, power, and energy and **evaluate** the importance of accuracy in measuring devices.	**K4**
4.	**Examine** the operation of transducers for measuring displacement, temperature and force.	**K3**
5.	**Test for** the dielectric strength of oil and **select** the oil based on quality.	**K4**

LIST OF EXPERIMENTS

1	Measurement of Power Using 3-Ammeter Method.
2	Measurement of Low Resistance using Kelvin Double bridge.
3	Measurement of Inductance using Anderson bridge.
4	Measurement of Inductance using Maxwell bridge.
5	Measurement of Capacitance using Schering bridge.
6	Calibration of Single-Phase Wattmeter.
7	Calibration of Single-Phase Energy Meter using Phantom Loading.
8	Measurement of Power using Two Wattmeter Method.
9	Testing of Dielectric Strength of Oil.
10	Study of LVDT and Capacitance Pickup-Characteristics and Calibration.
11	Resistance Strain Gauge-Strain Measurement and Calibration.
12	Calibration of Thermocouple and Thermister for temperature measurement.

Reference Books:

1.	Sawhney,.A.K., "A Course in Electrical & Electronic Measurement and Instrumentation," Dhanpat Rai & Company Private Limited, New Delhi, 18th Edition, 2007.
2.	Golding, E.W., and Widdis, F.C., "Electrical Measurements and Measuring Instruments," A H Wheeler & Com Company, Calcutta, 5th edition, 2003.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2205	**PC**	**--**	**--**	**3**	**1.5**	**15**	**35**	**3 Hrs.**

ELECTRICAL MACHINES- I LABORATORY

(For EEE)

Course Objectives: Students will Conduct

1.	Load test on DC shunt and DC series motors and Hopkinson's test.
2.	OC and SC tests and load test on single phase transformer.
3.	Swinburn's test and sumpner's test on DC shunt motor and transformer
4.	OCC of DC shunt generator and load test on DC compound generator
5.	Speed control of DC shunt motors.

Course Outcomes: Students will be able to

S.No	Outcome	Knowledge Level
1.	ConductLoad **tests** on DC motors and Hopkinson's **test** to **Analyze** the efficiency performance of DC Shunt and DC series motors.	K4, K5
2.	ConductOC and SC **tests** and Load **test** on transformer to **Analyze the** regulation and efficiency performance of the transformer.	K4, K5
3.	ConductSwinburn's **test** and Sumpner's**test** to **Predict** the Efficiency performance of Dc shunt machine and Transformer.	K4, K5
4.	ConductOCC on DC Shunt generator and Load **test** on DC compound generator to **Determine** the open circuit characteristics and over compound and under compound characteristics.	K4, K5
5.	Conduct a **test** onSpeed control of a DC shunt motor to **Analyze** the Speed characteristics of DC shunt motor.	K4, K5

LIST OF EXPERIMENTS

1.	Swinburn's Test
2.	Speed control of a DC shunt motor.
3.	Load test on DC Shunt motor.
4.	Load test on DC series motor.
5.	Load test on DC Compound generator.
6.	Open circuit characteristics of a DC shunt generator.
7.	Hopkinson's Test.
8.	Internal and external characteristics of a DC shunt generator.
9.	OC and SC tests on a single-phase transformer.
10.	Load test on a single-phase transformer.
11.	Sumpner's Test.

Reference Books:

1.	Electrical Machine by Dr. P. S. Bhimbra Khanna publications, 2011.
2.	Irving L. Kosow, "Electrical Machines & Transformers", Prentice Hall; 2nd Revised Edition 1990.
3.	Clayton, A.E; "Performance and Design of direct machines" CBS; IST edition (2004).

Code	Category	L	T	P	C	I.M	E.M	Exam
B20EE2206	**SOC**	**--**	**--**	**4**	**2**	**--**	**50**	**3Hrs.**

SMART SYSTEMS LABORATORY
(For EEE)

Course Objectives: Students will

1.	To study the various sensors used in IoT Networks, Raspberry Pi, Arduino and their peripherals,
2.	To study the wireless communication devices and their communication protocols used in IoT networks
3.	To study development of IoT application from a global perspective and building automation systems by considering real world constraints.
4.	To understand the implementation of IoT by studying case studies like Smart Home, Smart city, etc.

Course Outcomes: Student will be able to

S.no	Outcome	Knowledge Level
1.	Identity different components of IoT Networks	**K2**
2.	Interface various sensors to Processor boards	**K4**
3.	Apply various IoT network protocols to communicate between sensors and machines wirelessly	**K3**
4.	Connect the devices using web and internet in the IoT environment.	**K4**
5.	Develop automation of a system using IoT devices	**K4**

LIST OF EXPERIMENTS

1	Familiarization with Arduino/Raspberry Pi and perform necessary software installation.
2	Interfacing of 7 Segment Display/Touch Screen Display
3	Interfacing Analog Input & Digital Output boards
4	Interfacing and Connectivity of Various Sensor boards
5	To interface LED/Buzzer with Arduino/Raspberry Pi and write a program to turn ON LED for 1 sec after every 2 seconds.
6	To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection.
7	To interface a temperature sensor with Arduino/Raspberry Pi and write a program to print temperature and humidity readings.
8	To interface Bluetooth with Arduino/Raspberry Pi: i)write a program to send sensor data to a smartphone using Bluetooth. ii)write a program to turn LED ON/OFF when ‘1’/’0’ is received from a smartphone using Bluetooth.
9	Smart energy monitoring
10	Cloud based control of motor using MQTT protocol
11	Night Light Controlled System & IR Remote Control for Home Appliances
12	Fire Alarm Using Arduino

Reference Books:

1.	Vijay Madisetti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1 st Edition, VPT, 2014

2.	Simon Monk, “Programming the Raspberry Pi: Getting Started with Python”, January 2012, McGraw Hill ProfessionalMassimo Banzi, “Getting Started with Arduino”, First Edition, February 2009, O'Reilly Media, Inc
3.	Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things – Key Applications and Protocols”, Wiley, 2012.

Code	Category	L	T	P	C	I.M	E.M	Exam
B20MC2201	MC	2	--	--	--	--	--	--

ENGLISH PROFICIENCY

(Common to CE,EEE,ME,AIDS & CSBS)

Course Objectives: The students will be able to

1.	Communicate their ideas and views effectively
2.	Practice language skills and improve their language competency.
3.	Know and perform well in real life contexts
4.	Identify and examine their self-attributes which require improvement and motivation.
5.	Build confidence and overcome their inhibitions, stage freight, nervousness etc.,
6.	Improve their reading skills.

Course Outcomes: The students will

S.No	Outcome	Knowledge Level
1.	Improve speaking skills.	K3
2.	Enhance their listening capabilities	K3
3.	Learn and practice the skills of composition writing.	K3
4.	Enhance their reading and understanding of different texts.	K3
5.	Improve their communication both in formal and informal contexts.	K3
6.	Be confident in presentation skills.	K3

SYLLABUS

UNIT-I	**Listening Skills** Types of listening Hearing and Listening Listening as a receptive skill
UNIT-II	**Speaking Skills** Presentation skills Describing event/place/thing Extempore Debate Group Discussion
UNIT-III	**Reading Skills** Types of Reading (Intensive and Extensive reading, Skimming, Scanning) Reading/Summarizing News Paper Articles
UNIT-IV	**Writing Skills** Essay Writing (Argumentative, Analytical and Descriptive) E-Mail Writing Business Letters Resume Writing
UNIT-V	**Integrated Language Skills** Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking

Reference Books:	
1.	Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
2.	English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
3.	Enriching Speaking and Writing Skills, Orient Blackswan Publishers.
4.	The Oxford Guide to Writing and Speaking by John Seely OUP.
5.	Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw hill.
6.	Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
Note: Internal Assessment is carried out throughout the semester.	